Self-supervised confidence-aware denoising imputation of genomic data

SFU

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The problem

Most (cell type, assay) experiments were never run, and those that were differ in depth, read length, run type and platform. In the ENCODE Imputation Challenge, a baseline that averages each assay over other cell types beat all but two submitted methods [1].

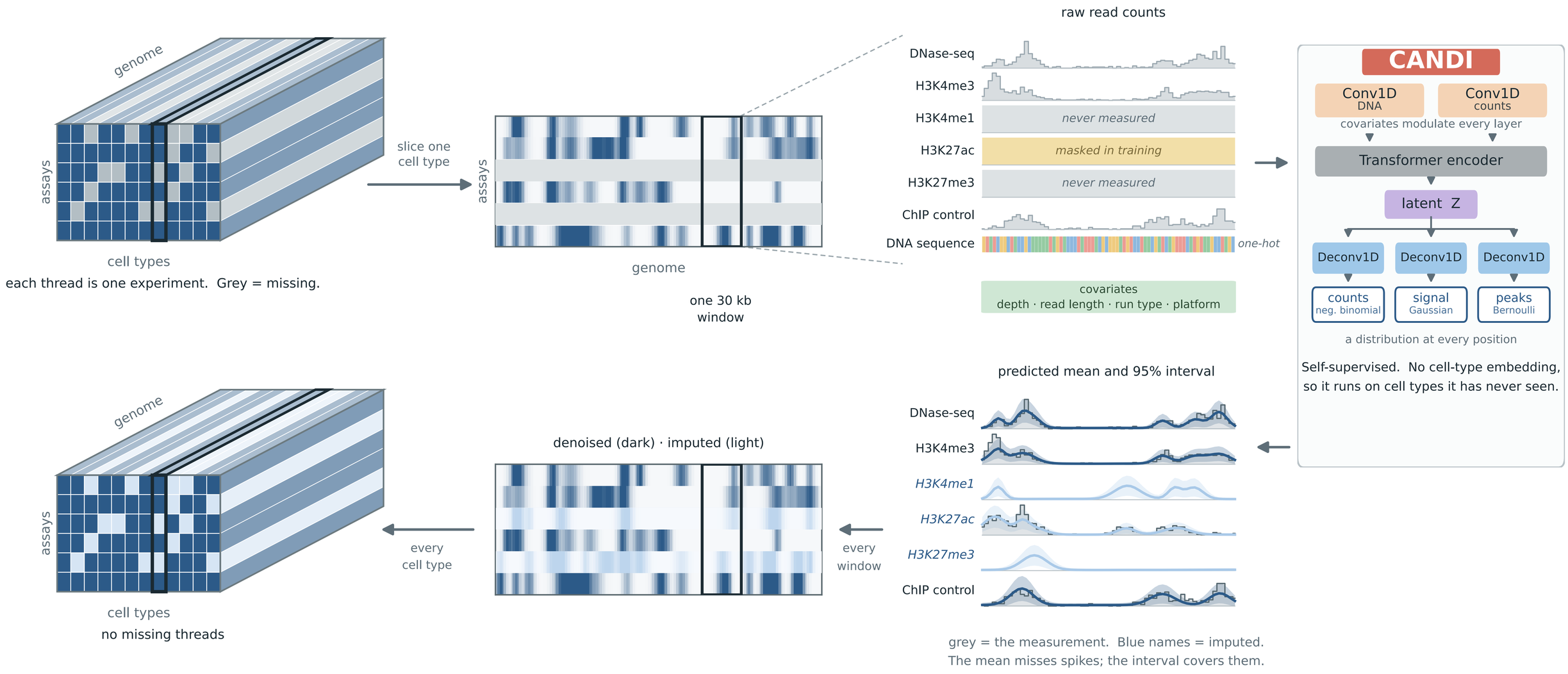
CANDI

Confidence-**A**ware **N**eural **D**enoising **I**mputer [2] is a self-supervised transformer over raw read counts, DNA sequence and four experimental covariates. It predicts a distribution for every assay in every 25 bp bin, so each imputed value has an interval. It has no cell-type embedding, so it runs on cell types it has never seen.

Key findings

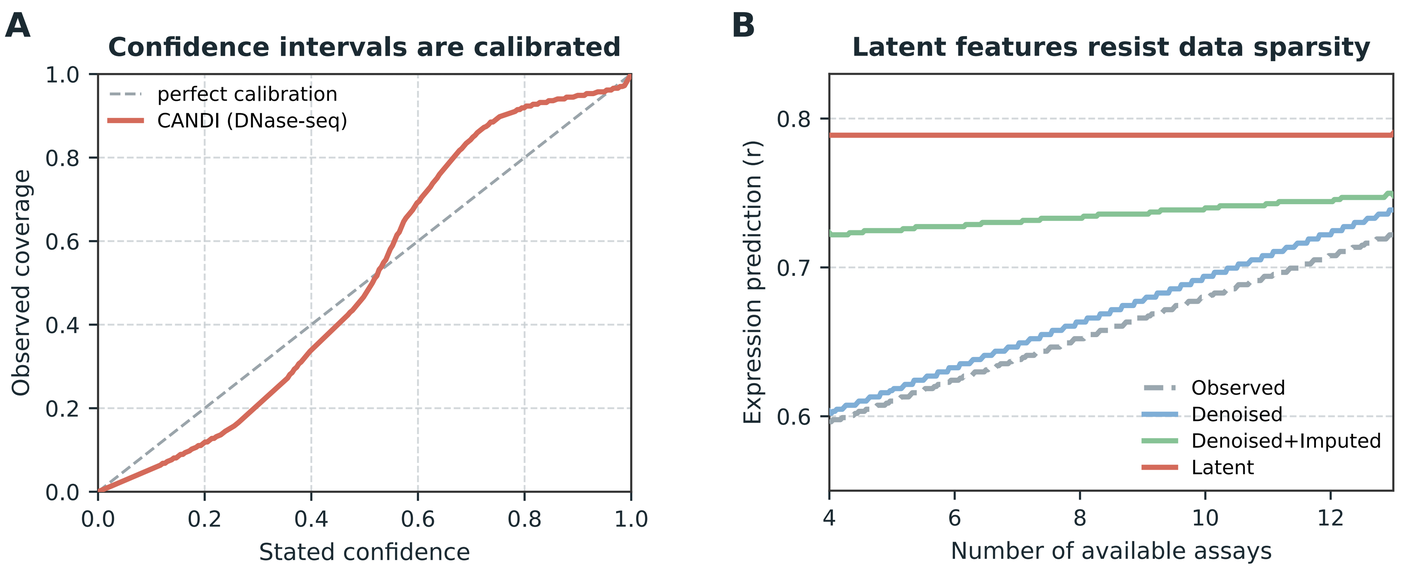
- CANDI is **1st of 26** on the ENCODE Imputation Challenge blind test.
- CANDI's intervals are calibrated, and its latent features are predictive of gene expression.

CANDI schematic workflow



Top: one cell type is sliced out of the cell type × assay × genome tensor, and a 30 kb window of it goes into CANDI as raw counts, DNA and covariates. Grey assays were never measured; yellow ones are masked in training. **Bottom:** CANDI returns a mean and a 95% interval for every assay, and window by window this fills every missing thread of the tensor.

Downstream utility of CANDI



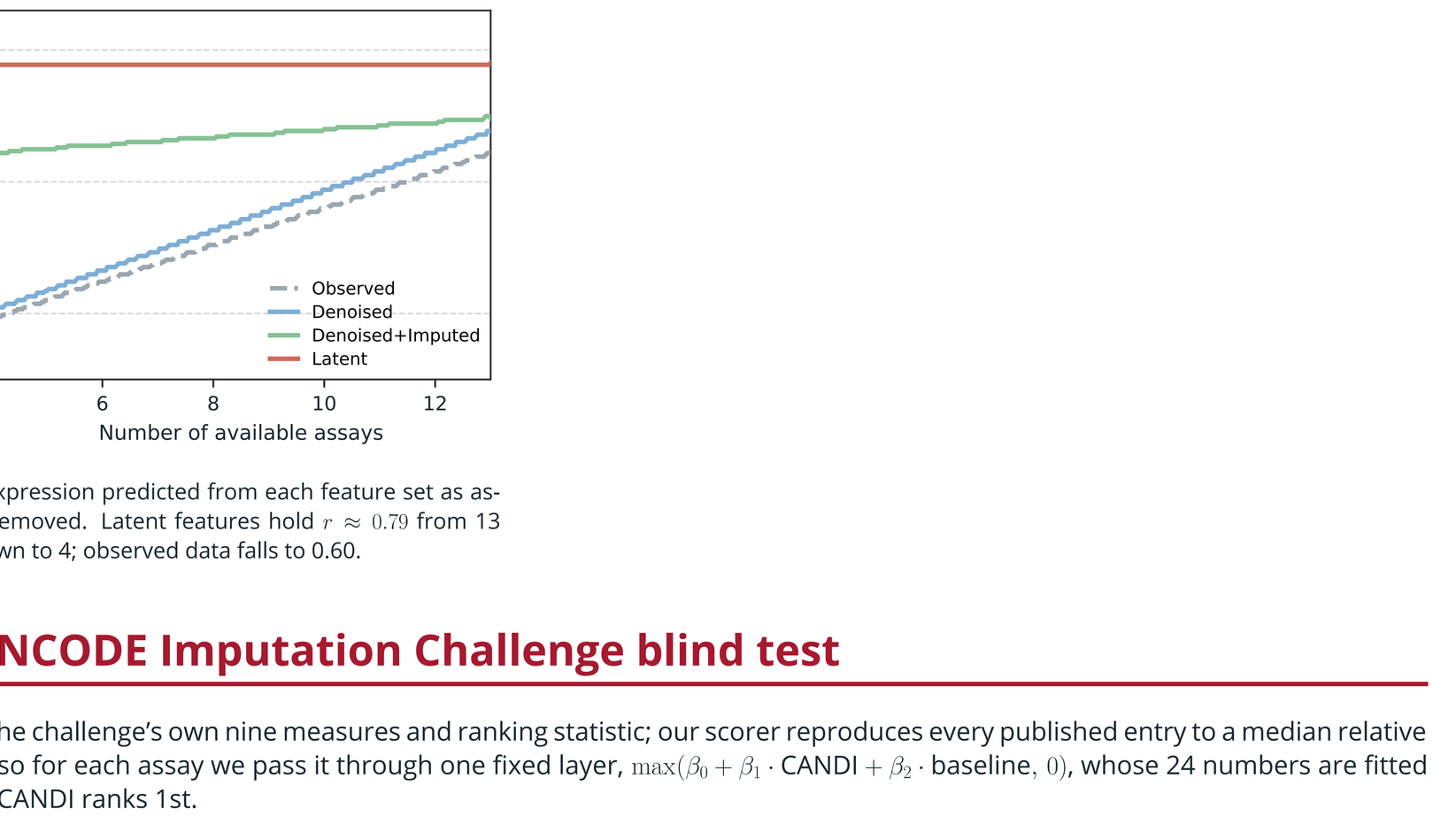
A For DNase-seq (cell line DND-41), the observed coverage of CANDI's intervals follows their stated confidence over the full range.

B Gene expression predicted from each feature set as assays are removed. Latent features hold $r \approx 0.79$ from 13 assays down to 4; observed data falls to 0.60.

Fixing the signal-level shift in the ENCODE Imputation Challenge blind test

51 blind-test experiments (12 cell types, 8 assays), scored with the challenge's own nine measures and ranking statistic; our scorer reproduces every published entry to a median relative difference of 9×10^{-10}. CANDI's output as it comes ranks 14th, so for each assay we pass it through one fixed layer, $\max(\beta_0 + \beta_1 \cdot \text{CANDI} + \beta_2 \cdot \text{baseline},\ 0)$, whose 24 numbers are fitted by least squares on the 133 training experiments only; with it, CANDI ranks 1st.

C The challenge's leaderboard with CANDI added. CANDI is first at 0.179 (bootstrap range 0.177–0.184); the next entry scores 0.306. Bars span the ten bootstrap groups.



D Each of the nine measures relative to the baseline, median over the 51 experiments. CANDI is first on genome-wide MSE, Pearson and Spearman, and in the top 8 on five more.

mehdiforoozandeh.github.io/CANDI

[1] Schreiber J, Boix C, *et al.* The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

[2] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels C–D: blind-test scoring from the Libbrecht lab's EIC reproduction (experiments 017, 025). *Baseline* is the challenge's average-activity track. The output correction was chosen after the challenge results were public; none of its numbers is fitted on blind-test data.